Emre Tiryaki

Final-Year Software Engineering Student



Samsun, Türkiye
+90 544 319 52 15
tiryakiemre18@gmail.com
emre-tiryaki.github.io
in/emre-tiryaki
emre-tiryaki

Professional Summary

Software engineer focused on building distributed systems, high-performance microservices architectures, and scalable backend infrastructures. Possesses foundational knowledge in Go; designs modern backend architectures using AWS, NoSQL, and in-memory databases. Develops low-latency, fault-tolerant services using CI/CD tools and DevOps practices. Aims to work on internal tools that automate operational processes.

Education

İnönü University — Malatya, Turkey
Software Engineering (B.Sc.) — *2023 – 2027*

Teknik Yetenekler

Programming Languages	TypeScript, JavaScript, Go, Rust
Backend	Gin, Express, Axum, Fastify, NestJS, GraphQL, REST API
Databases	I have hands on experience in both SQL and NoSQL databases
DevOps	Docker, Linux shell commands, Git, AWS(EC2, Lambda, S3), Even if I dont have any GCP experience, I am confident that I can adapt to it fast and smoothly

Experience

Avikon – Backend Developer Intern — August 2026 – October 2026

- Contributed to the development and maintenance of microservices-based backend services.
- Designed and implemented the GraphQL API layer using the Mercurius library.
- Built asynchronous inter-service messaging infrastructure using RabbitMQ.

Projects

Autonomous Finance Assistant (OFA) – Intelligent Payment Routing & Issue Resolution — July 2026
Moka United AI Ideathon & Hackathon — JavaScript, Express, Python, Machine Learning, React

- Developed a machine learning-based algorithm for selecting the optimal acquirer bank among multiple options.
- Designed an AI agent architecture that takes autonomous action when system issues occur without human intervention.

OKUR – AI-Powered Exam Reader — May 2026
İnüfest 2026 Competition — TypeScript, NestJS, Python, Machine Learning, React

- Developed a system that generates exam papers in LaTeX format.
- Built a module that processes exam papers using computer vision to extract and convert answers to text.
- Implemented an AI model that semantically compares student answers with the instructor's answer key for automated grading.

Cosmic Data Processing Pipeline – Big Data Processing Engine — March 2026
TUA Astro Hackathon — Python, FastAPI, React

- Developed a component that generates synthetic cosmic data and streams it to the processing system.
- Implemented an end-to-end data processing pipeline that performs real-time cleaning (DSP) of cosmic radiation-induced signal distortions using NumPy and SciPy vectorization, and autonomous anomaly detection with LSTM Autoencoder models achieving 90%+ accuracy.

HEGS News – Gamified News Platform — October 2025
Anadolu Ajansı Media Technologies Hackathon — JavaScript, Java, Spring Boot, Python, FastAPI

- Presented news data scraped from Anadolu Ajansı website through a clean user interface.
- Added a module enabling users to receive AI-generated summaries of current news.
- Developed gamification features including user profiles, friend additions, and mini-games aligned with the hackathon's "Gamifying News" theme.

Languages

Turkish native language
English Professional working proficiency

Independent Projects

- Developed Bash scripting skills on Arch Linux used as a daily driver operating system.
- Built `project-toolkit`, a bash script automating project creation and organization workflows.
- Developed `auto-readme-generator`, a tool that analyzes projects and automatically generates README files using user-specified AI models (Rust/Axum backend, React/JavaScript frontend).
- Created `pdf2markdown`, a Python command-line tool that converts PDF files to Markdown in parallel using models running locally via Ollama.
- Built `RepoGrab`, a TUI tool written in Go that enables downloading single files from GitHub and GitLab repositories without cloning the entire repository.